



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Build and Deploy Python Applications with Vibe Coding

TGS Ref No: TGS-2019504591

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.6

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	20 July 2026	Initial release — 5 topics, 20 labs, CardGuard fraud-screening application.	Dr. Alfred Ang
1.1	20 July 2026	Corrected all framing content to the Python / vibe-coding syllabus: learning outcomes, core-concepts section, trainer profile, lesson-plan schedule (9:30am-6:30pm), lab references (20 labs) and closing slides now render from course_data. Added Quick Command Reference and Support sections to the Learner Guide.	Dr. Alfred Ang
1.2	20 July 2026	Rebuilt the lab step slides: code now renders in a dark card sized to its content with a side-by-side explanation column, long snippets continue onto further slides, and the instruction text no longer overlaps the code. Added cover logos; replaced the exam-weighting kicker; Briefing for Assessment is now a visual tile grid.	Dr. Alfred Ang
1.3	20 July 2026	Fixed the lab code slides properly: code is wrapped to the card width so it can no longer render outside the dark card, and wrapped continuation lines keep their indentation instead of resetting to column 0. Lab kickers now show the lab number only (they were truncated mid-word). Replaced the LMS/TMS text link with a portal screenshot slide.	Dr. Alfred Ang
1.4	20 July 2026	Course-portal screenshot is now	Dr. Alfred Ang

		scaled to fit inside its card (it previously overhung the card and collided with the caption), and the caption sits below the card.	
1.5	20 July 2026	Re-authored Labs 1 and 2 onto the CardGuard application. They previously built an unrelated payroll/CPF tool while every other lab built CardGuard, so learners were told to work inside a payroll project in Lab 1 and inside the cardguard project from Lab 5. Lab 1 now initialises the cardguard project and loads transactions; Lab 2 teaches the same prompting pattern against a transaction risk-scoring function.	Dr. Alfred Ang
1.6	20 July 2026	Aligned the courseware with the new v17 assessment set: the practical instrument is a Case Study (CS, 80 minutes), not a Practical Performance (PP), and the WA is 50 minutes — corrected on the deck, the lesson plan and the learner guide. Code snippets now wrap on token boundaries so long pandas chains no longer split a string literal mid-word, and the explanation column no longer shows OOP notes beside Streamlit or Docker code.	Dr. Alfred Ang

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Course Information

Course Title	Build and Deploy Python Applications with Vibe Coding
WSQ Course Reference	TGS-2019504591
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	3 days · 8 training hours per day (24 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on labs using uv, an AI coding assistant, VS Code and Docker
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Apply vibe coding practices — set up a reproducible Python project with uv, and use an AI coding assistant to scaffold, explain and refactor working code.
- LO2: Design and implement object-oriented Python — classes, encapsulation, inheritance and composition — using AI-assisted conversational design.
- LO3: Build data analytics pipelines with pandas — load, clean, transform, group and aggregate realistic datasets into reportable results.
- LO4: Model and validate data with Pydantic and expose it through a FastAPI application with typed request and response models.
- LO5: Package and deploy a Python application — dependency locking, configuration, containerisation and a running deployed service.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 50 minutes, open book.
- Case Study (CS) — a CardGuard scenario with practical tasks drawn from the labs, 80 minutes, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 3 from 4:00 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Vibe Coding Foundations & Object-Oriented Python

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, ground rules and mandatory digital attendance (AM)
10:00–10:45	45 min	Topic 1 — Vibe Coding Foundations: uv Projects · AI

		Coding Assistants · Prompting Patterns · AI-Assisted Refactoring (concepts + demo)
10:45–11:00	15 min	Tea break
11:00–13:00	120 min	Hands-on: Lab 1: Set Up a Reproducible Python Project with uv; Lab 2: Write Effective Prompts for an AI Coding Assistant; Lab 3: Review and Correct AI-Generated Code
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Hands-on: Lab 4: Refactor Working Code Conversationally. Topic 2 — Object-Oriented Programming in Python: Classes · Encapsulation · Inheritance · Composition · Dunder Methods (concepts)
15:30–15:45	15 min	Tea break
15:45–18:15	150 min	Hands-on: Lab 5: Model a Domain with Classes and Dunder Methods; Lab 6: Encapsulate State with Properties and Validation; Lab 7: Build a Rule Hierarchy with Inheritance and Polymorphism; Lab 8: Compose Rules into an Engine
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Data Analytics with pandas, Pydantic & FastAPI

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–10:45	60 min	Topic 3 — Data Analytics with pandas: DataFrames · Cleaning · Transformation · GroupBy · Aggregation · Reporting (concepts + demo)
10:45–11:00	15 min	Tea break
11:00–13:00	120 min	Hands-on: Lab 9: Load and Explore Transactions with pandas; Lab 10: Handle Errors with Exceptions
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Hands-on: Lab 11: Build Per-Cardholder Baselines with GroupBy; Lab 12: Detect Velocity Bursts with Rolling Time Windows
15:30–15:45	15 min	Tea break
15:45–18:15	150 min	Topic 4 — Data Modelling with Pydantic and FastAPI: Typed

		Models · Validation · Endpoints · Request/Response Schemas · Auto Docs (concepts). Hands-on: Lab 13: Replace Hand-Written Validation with Pydantic Models; Lab 14: Expose the Screening Engine as a FastAPI Endpoint
18:15–18:30	15 min	Day 2 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 3 — FastAPI, Packaging, Deployment & Assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Hands-on: Lab 15: Persist Screening Decisions to SQLite; Lab 16: Test the API with pytest and httpx
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	Topic 5 — Packaging and Deployment: uv Lockfiles · Configuration · Containers · Deploying a Service (concepts). Hands-on: Lab 17: Build a Fraud Analyst Dashboard with Streamlit
13:00–14:00	60 min	Lunch break
14:00–15:05	65 min	Hands-on: Lab 18: Externalise Configuration and Protect Secrets; Lab 19: Lock Dependencies and Containerise the API; Lab 20: Deploy the Full Stack with Docker Compose
15:05–15:20	15 min	Tea break
15:20–15:35	15 min	Briefing for Assessment
15:35–16:25	50 min	Written Assessment (WA) — Short-Answer Questions (SAQ), 50 minutes, open book
16:25–17:45	80 min	Case Study (CS) — a CardGuard scenario with practical tasks drawn from the labs, 80 minutes, open book
17:45–18:30	45 min	TRAQOM survey, assessment digital attendance, sign Assessment Summary Record

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to exam skill areas)

Topic / Exam skill area	Weighting	Labs
Topic 01: Vibe Coding	15–20%	Lab 1, Lab 2, Lab 3, Lab 4

Foundations		
Topic 02: Object-Oriented Programming in Python	20–25%	Lab 5, Lab 6, Lab 7, Lab 8
Topic 03: Data Analytics with pandas	20–25%	Lab 9, Lab 10, Lab 11, Lab 12
Topic 04: Data Modelling with Pydantic and FastAPI	20–25%	Lab 13, Lab 14, Lab 15, Lab 16
Topic 05: Packaging and Deployment	15–20%	Lab 17, Lab 18, Lab 19, Lab 20